SWADHIKAAR (स्वाधिकार)

SOVEREIGN INDIC HEALTHCARE & EMERGENCY OPERATIONS PLATFORM

Four Doors. One Brain. Connects citizens, rural ASHA workers, hospital doctors, and 108 emergency dispatch into a single reactive clinical intelligence layer—powered by **Gnani.ai Prisma & Timbre**.

BEST CODE-MIXED VOICE · HINGLISH · TANGLISH · BENGLISH

GNANI PRISMA v3.0 (indic-auto) · TIMBRE v3.0

DPDP ACT 2023 COMPLIANT · 52 ROUTES

88% Cut
PHARMACY OUT-OF-POCKET SPEND
OCR salt matching: Telma 40 (₹285) → Generic (₹32)

< 3 Sec
VOICE INTAKE TO 108 CAD SBAR
Panicked vernacular speech to structured hospital handoff

52 Routes
PRODUCTION FULL-STACK PLATFORM
Next.js 16 + Supabase RLS across all 4 clinical surfaces

0% Drop
429 CONCURRENCY GUARD
FIFO serialization queue shields Gnani API rate limits

FOUR ACCESSIBLE CLIENT SURFACES · ONE SHARED CLINICAL CORE

DOOR 1 `/patient`

Citizen Caregiver PWA
Conversational symptom triage in Hinglish, Tanglish & Benglish, 3D WebGL shader orb, OCR & PM-JAY pharmacy savings.

DOOR 2 `/asha`

ASHA Field Portal
Offline-first field screening with IndexedDB local caching & auto-sync when cellular connectivity returns.

DOOR 3 `/ops`

Doctor Clinical Copilot
Real-time doctor workstation for hands-free clinical review, voice dictation, and acute bed-queue management.

DOOR 4 `/dispatch`

108 EMS Emergency CAD
Priority ambulance terminal with AVPU consciousness scale, live CPR timer, and instant structured SBAR payload.

SHARED CLINICAL CORE · REACTIVE BUS

FHIR R4 · SUPABASE REALTIME

- **Cross-Door Reactive State Sync:** Instant sync across citizen, ASHA, doctor, and dispatch via Supabase Realtime broadcast channels.
- **FHIR R4 Domain Schema:** Canonical Encounter, Observation, Condition, and SBAR clinical data structures with strict type contracts.
- **Deterministic NEWS2 Clinical Latch:** Standardized National Early Warning Score. Red-flag triggers (score ≥ 5) lock emergency state & auto-alert 108 dispatch.
- **Three-Tier PWA Data Plane:** Service Worker app shell caching, IndexedDB emergency facility storage, and network-first live voice streaming.

GNANI.AI SOVEREIGN VOICE STACK

CODE-MIXED INDIC CORE

- **Prisma v3.0 Indic Code-Mix STT:** Sub-200ms auto-detection of code-mixed **Hinglish** ("*chest heaviness*"), **Tanglish** ("*marbu vali & swasam kashtam*"), & **Benglish** ("*buke betha & saas koshto*") via `indic-auto`.
- **Timbre v3.0 Multilingual TTS:** 24kHz streamed vernacular audio with native personas: **Shivani** (Hinglish), **Lakshmi** (Tanglish), & **Riyaa** (Benglish) with natural ₹ currency cadence.
- **Evon v3.3 Triage Reasoning:** Indic clinical reasoning model with resilient multi-tier fallback (Gemini Flash & Groq Llama 3.3).
- **FIFO Concurrency Shield:** Promise serialization queue prevents 429 rate limit drops, guaranteeing 100% demo uptime under load.

EIGHT REALIZED INTERACTION TOPOLOGIES

T1 Citizen Intake: Multilingual symptom triage & scheme guidance.

T2 Proactive Care: Scheduled post-discharge follow-ups.

T3 Doctor Copilot: Voice dictation to structured EHR records.

T4 ASHA Copilot: Hands-free rural screening with offline sync.

T5 Emergency Lifeline: CPR guidance, 108 dispatch & SBAR handoff.

T6 Medical Bridge: 2-way vernacular clinician-patient interpretation.

T7 Vocal Indicators: Breath-interval Dyspnea Index tracking.

T8 LASA Safety: Phonetic check for Look-Alike Sound-Alike drugs.

CLINICAL SAFETY RAILS & ECONOMIC GROUNDING

Jan Aushadhi Substitution
Prescription OCR and fuzzy molecule matching against 1,800+ PM-JAY salts. Converts brand-name chronic prescriptions (e.g., Telma 40 ₹285) to generic equivalents (₹32), eliminating 88% out-of-pocket costs.

T7 Biomarkers & T8 LASA Guard
Real-time speech-pattern analysis tracks vocal breath intervals to detect acute respiratory distress. Phonetic distance verification prevents fatal Look-Alike Sound-Alike drug dispensing errors.

PM-JAY Hospital Readiness
Real-time directory of 1,800+ empanelled facilities with OSRM emergency routing, Ayushman Mitra pre-auth checklists, and live specialty bed verification to prevent critical golden-hour delays.

DESIGN INTENT: One shared clinical core across four role-specific doors. Clinical scoring and emergency escalation remain deterministic; Gnani.ai voice models provide the natural Indic language interface. The offline cache and simulated fallback paths ensure absolute operational resilience.

Live Platform: swadhikaar.vercel.app **Source Code:** github.com/ace-ify/swadhikaar **Compliance:** DPDP Act 2023 (Synthetic Clinical Telemetry)

THE GREAT INDIAN AI CHALLENGE · GNANI.AI